Allegri Compagni
HCI - 2023/24
Education and Learning
Assignment 1

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Domain of interest

Specific Domain: People who have had a gym membership (gym equipment room) for less than 6 months (beginners), between the ages of 18 and 30.

We chose this specific group because we believe there may be hidden needs among these individuals, regarding how to learn the basics about working out properly. Additionally, we focused on this specific age range to identify common needs among young people and narrow down the scope.

Methodology - Users

(2a Participants - Why were they chosen? Why are they appropriate? How were they recruited? Which is their age, gender, … ? Who is the extreme user and why? Who is the domain expert, if any?)

Immediate users (x4) : people between the ages of 21 and 26 that were chosen because of their little experience in the gym. Moreover, we looked for 2 males and 2 females to have a balanced point of view. They were recruited thanks to social media interactions, word of mouth.

Lead user (x1) : we chose a 25 years old guy with 6 years of experience in the gym because he could have already faced the same problems the immediate users are facing.

Domain expert (x1): we chose a 30 years old (male) personal trainer because he has a different point of view on the domain, that includes working with a lot of the immediate users we are considering. Also, he knows well the environment of the gym.

Methodology

Immediate Users - Questions:

(2c What did you observe (if any) and ask?)

- Why did you join the gym?
- What is your long-term goal?
- Do you engage in any other sports besides going to the gym? Which ones?
- What does it mean to you to know how to train?
- How did you feel during your first training sessions in the gym?
- How often do you go to the gym per week, and why?
- → Are there any specific days you prefer or prefer to avoid?
- What is the time of day when you usually train at the gym, and what factors do you consider when choosing it?
- How much time are you willing to dedicate to training?
- Do you prefer to train alone or with a partner? Why?
- What do you think of gym classes?
- What training methodologies do you use? (How do you structure your workouts?)
- How do you choose the exercises you do at the gym?
- → (optional) If you have a workout plan, who made it and how?
- → (optional) What resources did you use? Where did you get the necessary information?
- → (optional) Did they consider your starting level? In what way?
- If you have any pre-existing medical conditions, how do you manage them at the gym?
- Have you ever experienced any injuries or minor discomfort at the gym?
- → (optional) What caused it? What measures did you take to prevent it from happening again?
- How do you handle it if, for any reason, you are unable to perform a specific exercise?
- How has your training changed over time?
- How do you measure your progress at the gym?
- How do you seek to improve your knowledge of training?
- What were the top 3 difficulties you encountered during this initial period?
- Are you willing to spend additional money on your training? How?
- What are 3 things that you think could improve your gym experience? And 3 things that are currently working well?
- If you were to identify any mistakes you make during your workouts, what would they be?
- → How would you resolve them?

Methodology - Immediate Users Interviews

(2b Where were the interviews conducted?
2d Team members roles for each interview?
2e Any material you used (camera, recorder, etc.))

Immediate User 1 (female, 26yo)
The interview was conducted by Lisa Cangialosi, who took notes, and Federico Castriotta, who asked the questions. The participant was Sara, a 26 years old girl that joined the gym 5 months ago. She was chosen as an immediate user for her little experience in the gym. She was recruited through a friend of one of the team members. We interviewed her while she was at home through video call on Whatsapp, we took written notes and we recorded the interview.

Immediate User 2 (male, 22yo)
The interview was conducted by Marco Donnarumma, who took notes, and Luca Filippetti, who asked the questions. The participant was Claudio, a 22 years old guy that joined the gym 4 months ago. He was chosen as an immediate user for his little experience in the gym. He was recruited since he's a friend of one of the team members. We interviewed him in a bar next to his gym just after a training session, we took written notes and we recorded the interview.

Methodology - Immediate Users Interviews

(2b Where were the interviews conducted?
2d Team members roles for each interview?
2e Any material you used (camera, recorder, etc.))

Immediate User 3 (female, 23yo)
The interview was conducted by Luca Filippetti, who took notes, and Marco Donnarumma, who asked the questions. The participant was Jimena, a 23 years old girl that joined the gym 2 months ago. She was chosen as an immediate user for his little experience in the gym. She was recruited through direct acquaintance with a team member. We interviewed her through video call, we took written notes and we recorded the interview.

Immediate User 4 (male, 21yo)
The interview was conducted by Federico Castriotta, who took notes, and Lisa Cangialosi, who asked the questions. The participant was Nicolo, a 21 years old guy that joined the gym 3 months ago. He was chosen as an immediate user for his little experience in the gym. He was recruited through social media interaction. We interviewed him at a bar, we took written notes and we recorded the interview.

Methodology

Lead User - Questions:

(2c What did you observe (if any) and ask?)

- Why did you join the gym?
- What is your long-term goal?
- Do you practice any other sports besides going to the gym? Which ones?
- How long have you been going to the gym?
- What does it mean to you to know how to train?
- → How did you learn to train better?
- How often do you go to the gym per week, and why?
- → Are there any specific days you prefer or prefer to avoid?
- What is the time of day when you usually train at the gym, and what factors do you consider when choosing it?
- How much time are you willing to dedicate to training?
- Do you prefer to train alone or with a partner? Why?
- What do you think of gym classes?
- What training methodologies do you use? (How do you structure your workouts?)
- → Has it changed over time?
- How do you choose the exercises you do at the gym?
- → (optional) If you have a workout plan, who made it and how?
- → (optional) What resources did you use? Where did you get the necessary information?
- → (optional) Did they consider your starting level? In what way?
- If you have any pre-existing medical conditions, how do you manage them at the gym?
- Have you ever experienced any injuries or minor discomfort at the gym?
- → (optional) What caused it? What measures did you take to prevent it from happening again?
- How do you handle it if, for any reason, you are unable to perform a specific exercise?
- How has your training changed over time?
- How do you measure your progress at the gym?
- How do you seek to improve your knowledge of training?
- Are you willing to spend additional money on your training? How?
- What are 3 things that you think could improve your gym experience? And 3 things that are currently working well?
- If you were to identify any mistakes you make during your workouts, what would they be?
- → How would you resolve them?

Methodology - Lead User Interview

(2b Where were the interviews conducted?
2d Team members roles for each interview?
2e Any material you used (camera, recorder, etc.))

Lead User 1 (male, 25yo)

The interview was conducted by Luca Filippetti, who took notes, and Marco Donnarumma, who asked the questions. The participant was Mnar, a 25 years old guy that joined the gym 6 years ago. He was chosen as a lead user due to his large experience in the gym. He goes to the same gym as one of the team members, so he kindly participated to our activity. We interviewed him at Polytechnic, we took written notes and we recorded the interview.

Methodology

Domain expert - Questions:

(2c What did you observe (if any) and ask?)

- How long have you been a personal trainer?
- What did you study to carry out your profession?
- What does it mean to you to know how to train?
- What does your job entail?
- How many clients do you work with?
- → How many clients do you work with in a single day?
- → Are these private sessions?
- What are 3 positive aspects of working out with a personal trainer, and why would you recommend it to a new member?
- What do you think are 3 difficulties faced by a new gym member?
- What training methodologies do you use with your clients? (Only machines? Bodyweight alternatives?)
- → What works best for new members?
- How important is their starting level on a scale of 1 to 5 (where 1 is very low and 5 is very high)?
- → How do you take it into account?
- How do you handle it if one of your clients has a pre-existing medical condition?
- In the case of an injury to one of your clients at the gym, how do you handle it?
- How does the training you provide to your clients evolve over time?
- What are 3 pieces of advice for a new gym member, and 3 things that you think they may be doing wrong?
- What are 3 things you enjoy about your job, and 3 areas you think could be improved?
- Is there any important question that we haven't asked you?

Methodology - Domain Expert Interview

(2b Where were the interviews conducted?
2d Team members roles for each interview?
2e Any material you used (camera, recorder, etc.))

Domain expert 1 (male, 30yo)

The interview was conducted by Lisa Cangialosi, who took notes, and Federico Castriotta, who asked the questions. The participant was Alessandro, a 30 years old guy that has been working as a personal trainer for 5 years. He was chosen as a domain expert due to his large experience with young people approaching the gym. He was recruited through direct acquaintance with a team member. We interviewed him through video call, we took written notes and we recorded the interview.

Results

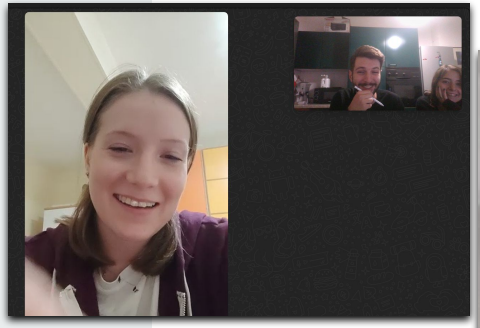
(3a. Pictures and relevant artifacts
3b. Key quotes)

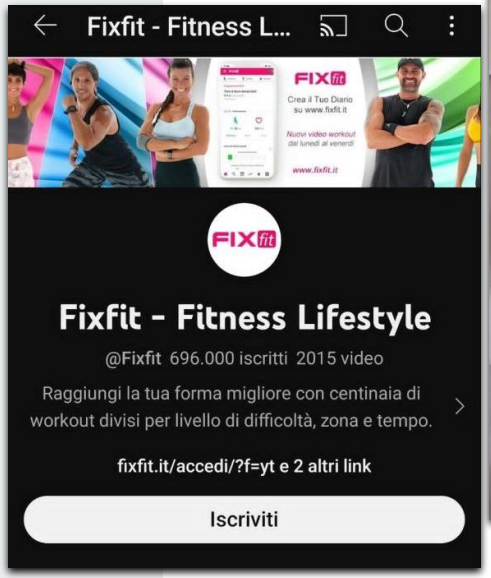
"... Knowing how to train means having a workout plan."

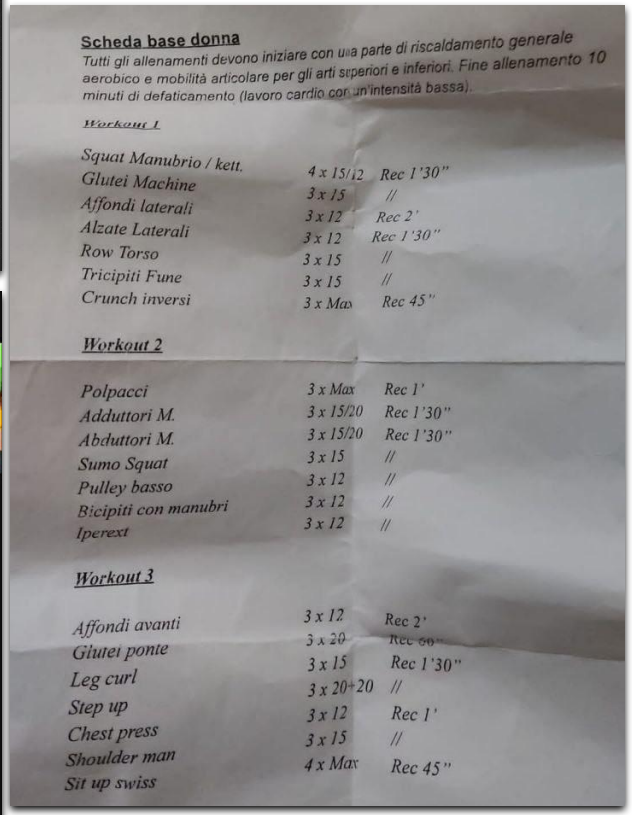
"... I added stretching to my workout plan... I looked for video tutorials on the internet "

"I don't take the time to properly learn how to use a specific piece of equipment, so I don't perform the exercise correctly. Not always instructors are available... Therefore, I usually do everything on my own."

Immediate User 1 (female, 26yo)







Scheda base donna

Tutti gli allenamenti devono iniziare con una parte di riscaldamento generale aerobico e mobilità articolare per gli arti superiori e inferiori. Fine allenamento 10 minuti di defaticamento (lavoro cardio con un'intensità bassa).

Workout 1

Squat Manubrio / kett.	4 x 15/12	Rec 1'30"
Glutei Machine	3 x 15	//
Affondi laterali	3 x 12	Rec 2'
Alzate Laterali	3 x 12	Rec 1'30"
Row Torso	3 x 15	//
Tricipiti Fune	3 x 15	//
Crunch inversi	3 x Max	Rec 45"

Workout 2

Polpacci	3 x Max	Rec 1'
Adduttori M.	3 x 15/20	Rec 1'30"
Abduttori M.	3 x 15/20	Rec 1'30"
Sumo Squat	3 x 15	//
Pulley basso	3 x 12	//
Bicipiti con manubri	3 x 12	//
Iperext	3 x 12	//

Workout 3

Affondi avanti	3 x 12	Rec 2'
Glutei ponte	3 x 20	Rec 60"
Leg curl	3 x 15	Rec 1'30"
Step up	3 x 20+20	//
Chest press	3 x 12	Rec 1'
Shoulder man	3 x 15	//
Sit up swiss	4 x Max	Rec 45"

Results

(3a. Pictures and relevant artifacts
3b. Key quotes)

“I didn't know where to start, I didn't know how to perform the exact movements of the exercises.”

“At the beginning, my gym provides pre-printed workout plans that do not consider the starting level and, in my opinion, are pointless.”

“The major difficulties were: understanding how to use the machine correctly, figuring out which machines to use for training a specific muscle group, and how to combine exercises for different muscle groups…”

Immediate User 2 (male, 22yo)

Immediate User 3 (female, 23yo)

Results

(3a. Pictures and relevant artifacts
3b. Key quotes)

"Training properly means not getting injured."

"I believe that training with a partner is better, but finding the right partner is difficult due to different goals and scheduling."

"I mainly get information online through YouTube videos and influencers."

"One of the main issues is consistency..."

Immediate User 4 (male, 21yo)

Results

(3a. Pictures and relevant artifacts
3b. Key quotes)

"In the beginning, it felt like a burden, I didn't have the motivation... but with a gym buddy, it's more enjoyable and convenient to push yourself to failure during sets."

"Knowing how to train means working out with the right technique."

"If a machine is occupied, I avoid substituting it with another because each machine targets a specific muscle group."

"To perform the exercises correctly, I have watched many videos on YouTube and researched which exercises target each muscle group."

GIORNO 1

panca inclinata 4x8 riposo 1.30 min vuoto
alzate barra sopra la testa 4x8 riposo 1.30 min vuoto
panca piana con manubri 4x10 riposo 1 min 8kg
alzate laterali 4x10 1 min 4kg
cavi per petto 4x8 1 min
2.5kg
squat 4x6 1.30 min vuoto
Lungees 3×16 1 min
Polpacci 4×10 1 min
Circuito addominali
GIORNO 4

trazione aiutate 3x5, 1 min 21kg
panca piana 4x8, 1.30 min
Spinte con manubri panca 30 grandi 4×10
Chest supported low row (Panca a 45 gradi con manubri) 4x8, 1.30 min
alzate sopra la testa con i manubri 4x8, 1 min
alzate laterali 4x10, 1 min
bicipiti con barra a curvette 4x10, 1 min
bicipiti su panca inclinata 4x10, 1 min
tricipiti skull crushers (barra piegata su panca, porti la barra dietro la testa) 4x10, 1 min
dips aiutati, 3x(esaurimento), 1 min

GIORNO 2

rematore (prendi la barra da terra e la porti all'ombelico) 4x8 1.30
Shrugs 4×8 1 min
pulley 4x8 1min 25kg
face pulls (prendi la corda e la porti alla faccia) 15kg 5x(15,12,12,10,10) riposo 1min
lat pulldown 4x8 1.30 30kg
SUPERSET
12 bicipiti ai cavi 7.5kg
12 tricipiti ai cavi 10kg
riposo 1 min alla fine di entrambi gli es
tutto x4

Results

(3a. Pictures and relevant artifacts
3b. Key quotes)

"For me, knowing how to train means doing exercises at the right time, with adequate recovery, proper nutrition, rest, and maintaining a healthy weight."

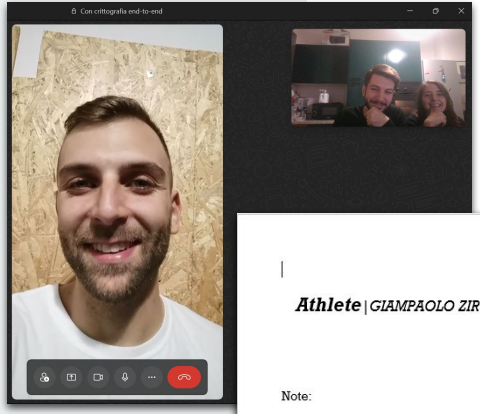
"I prefer to train with a partner if they are aligned with me in terms of experience. It is helpful for spotting. "

"Before, I used to have incorrect technique by overloading the machine, and I would train without a well-detailed program."

Lead User 1 (male, 25yo)

Results

(3a. Pictures and relevant artifacts
3b. Key quotes)



Domain expert 1 (male, 30yo)

Athlete | *GIAMPAOLO ZIRONE*

WEEK 1

| *Strength and Conditioning Training*

Note:

DAY 1	DAY 2	DAY 3
Warm Up	Warm Up	Warm Up
EMOM 15' • Minuto 1: max rep jumping jacks • Minuto 2': max rep air squat • Minuto 3': max rep push-up • Minuto 4': max rep burpees • Minuto 5' : REST	AMRAP 15' (AS MANY REP AS POSSIBLE) 5 BURPEES 10 CRUNCH 15 SQUAT	21-15-9 RESP OF LUNGEES ALT DIPS SULLA SEDIA V-UP ALTERNATI

RICORDATI DI FARE ESERCIZI SULLA FIT BALL + QUESTI ESERCIZI DI STRETCHING, 30'' PER OGNI POSTAZIONE

Cobra
cat positio
Psoas
child posit
postura la muro

https://www.youtube.com/watch?v=dierbrzmc1y
CTRL+clic per aprire collegam.

"Knowing how to train means making your workouts more effective in the least amount of time possible."

"The starting point is fundamental; starting from there, you build everything else."

"Even after many years of study, what personal trainers say is overshadowed by the rumors that come from social media and the media."

"Everything that can be done with a machine can be done with bodyweight, with or without weights."

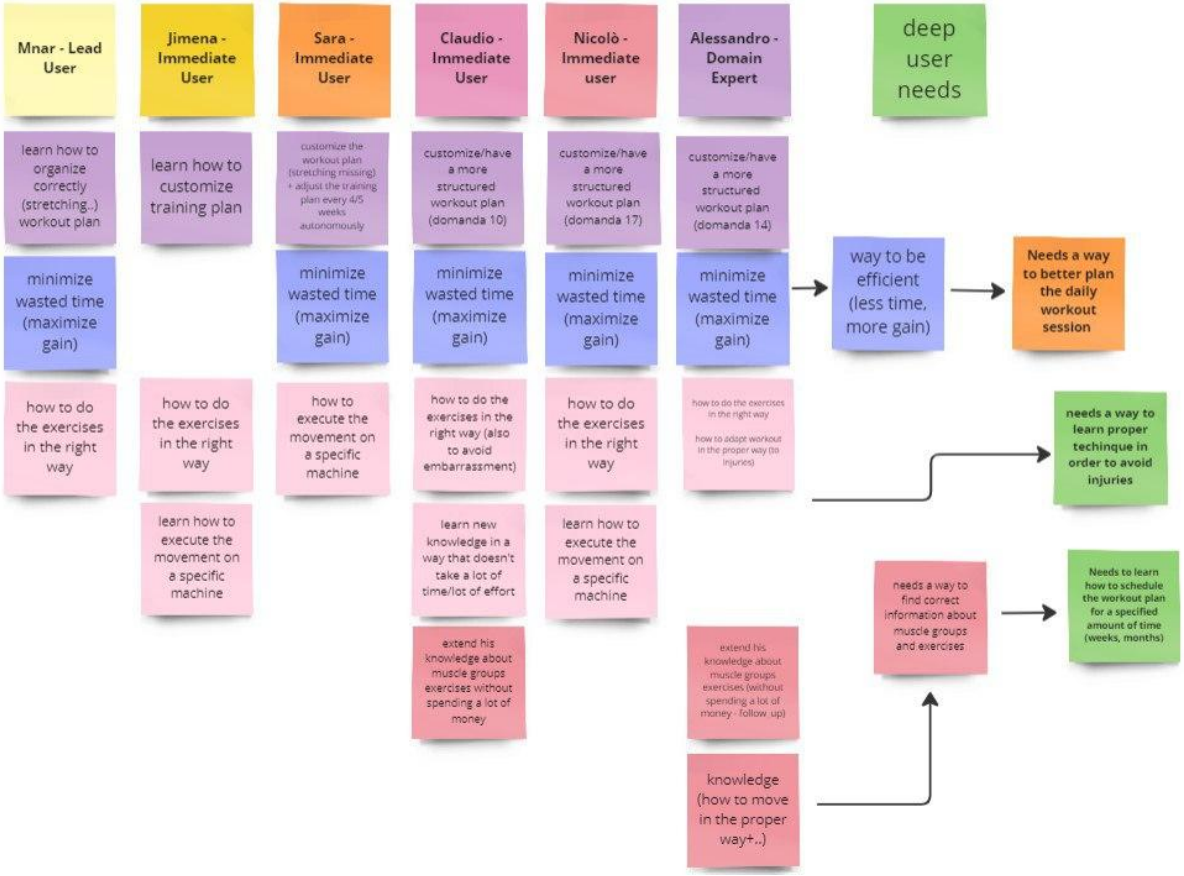
User needs

(4a. Include a list/picture of all brainstormed user needs)

We collected ideas from many interviews and grouped similar problems to find common issues shared by different users.

After that, we looked at the most common problems and tried to figure out what users might need.

At first, we identified numerous user needs (showed in orange and green in the picture on the side). We then narrowed down our focus and selected three deep user needs.

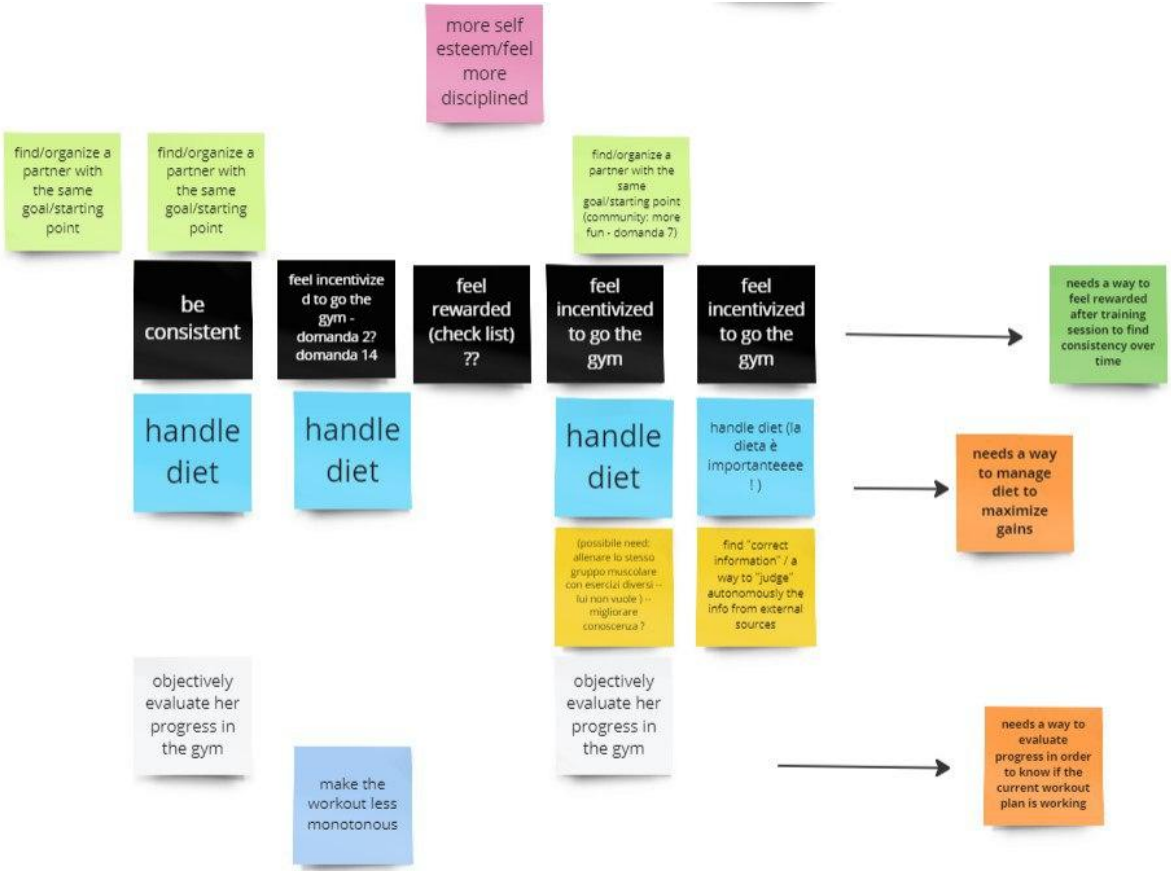


User needs

(4a. Include a list/picture of all brainstormed user needs)

The three deep user needs that we extracted were:

- "The user needs a way to learn proper exercise technique in order to avoid injuries"
- "Needs to learn how to schedule the workout plan for a specified amount of time (weeks, months)"
- "The user needs a way to feel rewarded after training session to find consistency over time."



Deep user need - 1

(4b. Present the 3-4 deep user needs; connect each of them to one or more interviews and answers)

“The user needs a way to learn proper exercise technique in order to avoid injuries”

“You have to know your body and your limits, you have to learn how to stop or modify your training according to your physique” (**Domain expert 1**)

“...I had shoulder problems but I didn't want to stop training, I had to adapt exercises with respect to that.” (**Lead user 1**)

“I don't take the time to properly learn how to use a specific piece of equipment, so I don't perform the exercise correctly. Not always instructors are available... Therefore, I usually do everything on my own.” (**Immediate user 1**)

“During the last training sessions I felt muscle discomfort without understanding the cause” (**Immediate user 2**)

“Training properly means not getting injured.” (**Immediate user 3**)

“Knowing how to train means working out with the right technique.” (**Immediate user 4**)

Deep user need - 2

(4b. Present the 3-4 deep user needs; connect each of them to one or more interviews and answers)

“Needs to learn how to schedule the workout plan for a specified amount of time (weeks, months)”

“... Knowing how to train means having a workout plan.”
(**Immediate user 1**)

“At the beginning, my gym provides pre-printed workout plans that do not consider the starting level and, in my opinion, are pointless.”
(**Immediate user 2**)

“The major difficulties were: understanding how to use the machine correctly, figuring out which machines to use for training a specific muscle group, and how to combine exercises for different muscle groups...”
(**Immediate user 2**)

"Initially, I lacked organization, but now I attempt to plan my daily workouts around specific muscle groups. However, it remains challenging due to my limited experience."
(**Immediate user 3**)

"Currently, I create my workout plans independently, but this wasn't the case when I first started. At the beginning of my fitness journey, I relied on online resources to learn how to design a workout plan. However, I encountered several challenges with this approach, such as information overload and the need to distinguish between reliable and unreliable sources."
(**Lead user 1**)

Deep user need - 3

(4b. Present the 3-4 deep user needs; connect each of them to one or more interviews and answers)

"The user needs a way to feel rewarded after training session to find consistency over time."

"It is very common for people to stop training at the first problem; instead, it is important to understand that training is adaptive and consistency is important" (**Domain expert 1**)

"Going to the gym spurs a lot more than working out at home: seeing other people working out work out makes you have more incentive to commit and be consistent…" (**Immediate user 1**)

"...I was having a hard time finding constancy and the right motivation, going to the gym just to put x in the checklist…" (**Immediate user 2**)

"One of the main issues is consistency..." (**Immediate user 3**)

"In the beginning, it felt like a burden, I didn't have the motivation... but with a gym buddy, it's more enjoyable and convenient to push yourself to failure during sets." (**Immediate user 4**)

Solutions

(5a. Include some pictures/screenshots of your brainstorms)

For every deep user need, we found some possible solutions, as it can be seen in the images on the side

needs a way to learn proper techinque in order to avoid injuries

- interactive multimedia content to show proper exercise technique and muscle groups used in it
- private lesson with a personal trainer
- Partner with a more expert workout buddy at the gym
- ask gym staff for guidance
- start with lighter weights to focus on form before progressing to heavier weights

Needs to learn how to schedule the workout plan for a specified amount of time (weeks, months)

- interactive multimedia content on scheduling principles
- pre-defined workout plans
- workout plan made by personal trainer
- fitness community to ask questions and share experiences

needs a way to feel rewarded after training session to find consistency over time

- gamification to incetivize the user in order to make him/her feel rewarded
- social space to share your progresses
- objectively evaluate progress
- a partner with the same goal
- discounts for gym materials

The chosen solution

(5b. Describe the solution you choose in 1-3 sentences; connect it with the need(s) it aims to solve
5c. How and why did you selected it?)

The final solution involves interactive multimedia content to show proper exercise technique and muscle groups used in it

The proposed solution aims to cover the following need:

- ***"needs a way to learn proper techinque in order to avoid injuries"***

We chose this solution because according to the majority of our team it best fits the target domain and their lack of experience.

Project name and value proposition

(6a. How did the team land on the name?
6b. One-line value proposition)

Gym Tonic

“Get toned up with fun”

This naming decision was guided by several key reasons that emphasize its significance and relevance:

- Clarity and Relevance: "Gym Tonic" immediately provides clarity about the context of our project, implying a focus on the world of fitness.
- Creative Wordplay: The name "Gym Tonic" incorporates clever and creative wordplay that echoes the famous cocktail "Gin Tonic." This connection suggests an intrinsic link between physical well-being and enjoyment.